

Trupti Kularkar

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PROFILE

Data professional with 1.8+ years applying statistical modeling, generative AI, and machine learning to solve high-dimensionality business problems in production environments. Experience spans anomaly detection (BigQuery ML, adaptive statistical thresholds), predictive data-quality frameworks, and LLM-based agentic systems — with a track record of translating analytical findings into measurable business outcomes (~92% detection accuracy, ~35% incident reduction, ~75% autonomous resolution).

TECHNICAL SKILLS

Data Analysis & Statistics: Statistical Analysis, Hypothesis Testing, Anomaly Detection, Predictive Thresholding (BigQuery ML), Trend Analysis, Backtesting

Generative AI/ML: RAG, hybrid retrieval (FAISS + BM25 + Reciprocal Rank Fusion), prompt & context engineering, agentic workflows, LLM orchestration, Vertex AI, Gemini API

Language: Python, SQL (DDL, DML, query & performance tuning)

Data Engineering & ETL: ETL/ELT Pipeline Design, Data Warehousing, Data Quality Frameworks, Apache Airflow, BigQuery

Cloud Platform (GCP): BigQuery, Cloud Storage, Pub/Sub, Cloud Functions, Cloud Run, Dataflow — transferable to AWS (Redshift, Glue, Lambda)

EXPERIENCE

Programmer Analyst, Cognizant Technology Solutions 11/2025 – Present | Pune, India

- Built an **LLM-powered ticket resolution agent** on Vertex AI (Gemini 2.5 Pro) that automated ~60% of repetitive support tickets, cutting mean resolution time from 4 hours to under 30 minutes.
- Designed SQL-based data validation frameworks across 15+ ETL pipelines, enforcing row-count reconciliation, null checks, duplicate detection, and schema consistency — reducing data quality incidents by ~35%.
- Optimized BigQuery query costs by ~25% via partitioning and clustering; automated 20+ Airflow DAGs, reducing manual operational overhead by 40% and cutting pipeline failure response time by 50%.
- Architected event-driven, real-time ETL pipelines (Cloud Storage → Pub/Sub → Cloud Functions → BigQuery) processing 500K+ daily events with sub-2-minute end-to-end latency.
- Collaborated with cross-functional teams to integrate AI-based monitoring into enterprise workflow systems.
- Mentored junior engineers on GCP tooling and Airflow pipeline development.

Programmer Analyst Trainee, Cognizant Technology Solutions 11/2024 – 11/2025 | Chennai, India

- Gained hands-on experience in GCP, Python, SQL, and ETL development.
- Built a **retail data pipeline** for sales trend analysis and forecasting using structured datasets.

PROJECTS

AI-Driven Data Quality Monitoring & Anomaly Detection, BigQuery ML, SQL, Gemini 2.5 Pro

- Built an **AI-driven anomaly detection** system using **BigQuery ML** Linear Regression with IQR-based adaptive thresholds — replacing hardcoded rules and improving anomaly detection accuracy to ~92% with a false positive rate under 5% across 6 months of backtested data.
- Integrated **Gemini 2.5 Pro** via **Vertex AI** for automated **root-cause analysis** on flagged anomalies, reducing mean investigation time from ~45 minutes to under 5 minutes and enabling **real-time data quality** monitoring at scale.

Local RAG & Hybrid Retrieval Sandbox, Python, FAISS, BM25, Google Gemini API, Ollam

- Built a locally deployable RAG playground combining dense (FAISS) and sparse (BM25) retrieval via Reciprocal Rank Fusion, with a configurable chunking engine and visual pipeline trace for inspecting ranks and relevance scores.
- Architected a provider-agnostic generation layer supporting Gemini API, local Ollama models, and offline simulation — enabling LLM orchestration testing without recurring cloud costs.

AI AutoHeal Agent – Automated Job Failure Remediation, GCP Cloud Run, BigQuery, Airflow, Gemini 2.5 Pro, Python, SQL

- Architected an autonomous **AI agent** on **GCP Cloud Run** that detects **Airflow DAG** failures from **BigQuery** and auto-remediates recoverable issues using **Gemini 2.5 Pro** classification — resolving ~75% of transient failures without human intervention and cutting on-call response time by ~60%.
- Implemented a **data quality** guardrail to block auto-remediation for integrity-related failures, preventing incorrect fixes; built end-to-end audit tracking with **NDJSON** logs loaded into **BigQuery**, achieving 100% job-level traceability across all remediation events.

Automated Sales Data Validation DAG, Apache Airflow, BigQuery, SQL, Python, Power Automate

- Engineered a configurable **Apache Airflow data pipeline** to automate sales **data validation** across source and target systems using CTE-based **SQL** variance logic and **BigQuery** audit logging — cutting manual validation effort by ~70% and catching data discrepancies within minutes instead of hours.
- Implemented dynamic runtime configuration via **dag_run.conf**, enabling zero-code onboarding of new validation workflows and reducing pipeline setup time from days to under 30 minutes; integrated **real-time alerts** via Microsoft Teams using **Power Automate REST API**.

EDUCATION

B.Tech (Artificial Intelligence), G. H. Raisoni College of Engineering, Nagpur 2020 – 2024 | Nagpur, India
CGPA: 8.83

CERTIFICATES

Google Cloud Certified Associate Data Practitioner, Discover Data Analysis Badge by Microsoft

PUBLICATIONS

Intelligent Diabetes Predicting Model for Diverse Ethnicities, Springer Nature Singapore